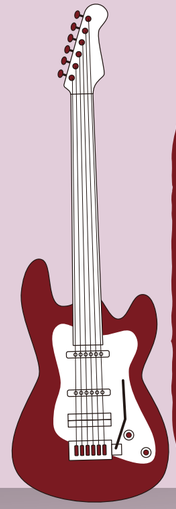
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Song Lyrics Mood Classification

@reallygreatsite

EMOLYRICS

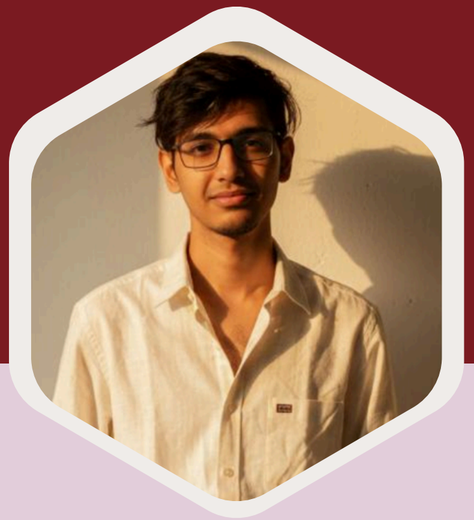


Rhythm

Culture

Emotion

OUR BEST TEAM



Harshit Raj 403



Sristi Priya 399



Chandrima Chawdhury 359



Dibyendu Khamaru 412

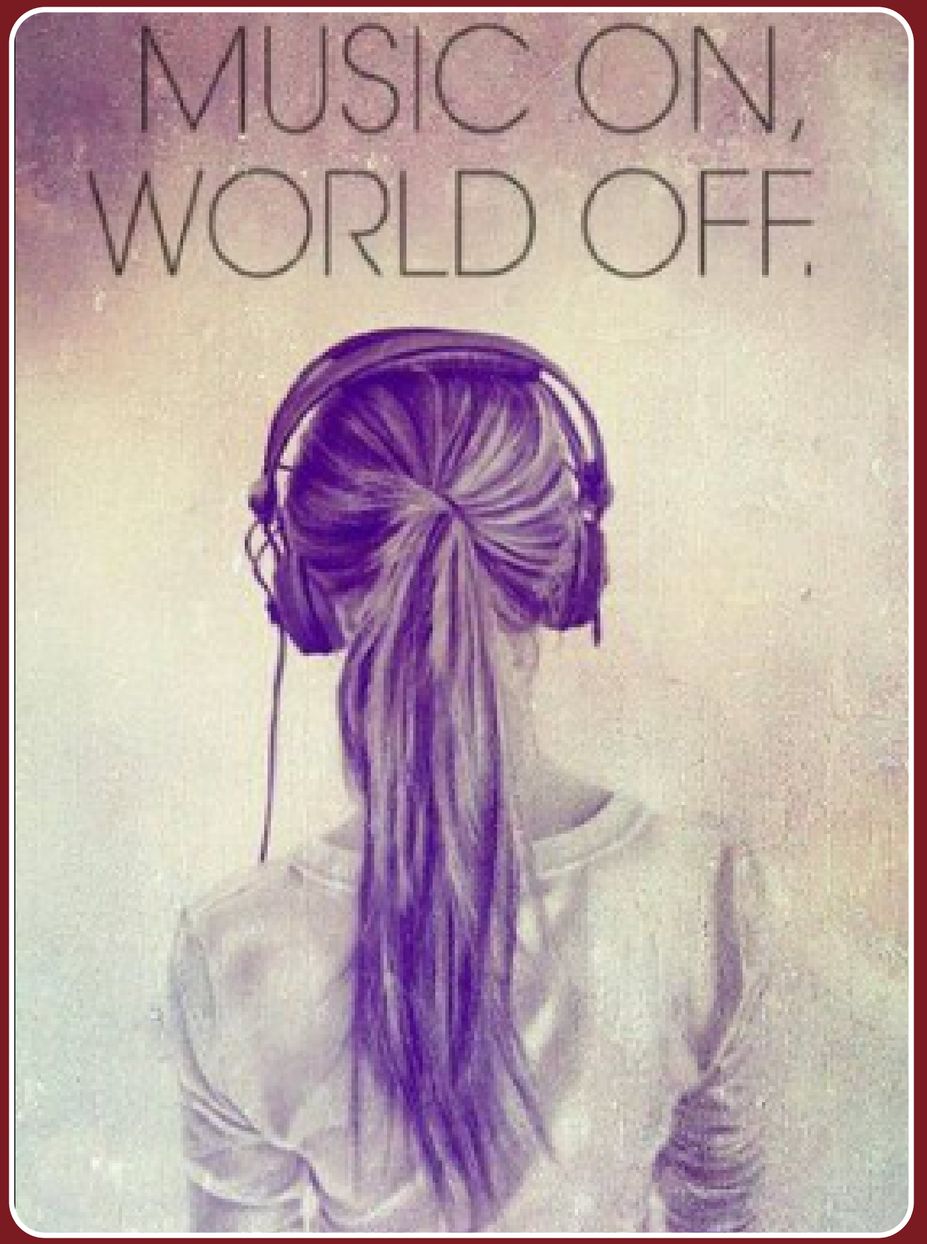


Tribeni Ghorai 455

PROBLEM STATEMENT

Understanding the emotional tone of song lyrics can enhance music recommendation systems. This project aims to classify song lyrics into different mood categories based on textual analysis.





OBJECTIVE OF THE PROJECT



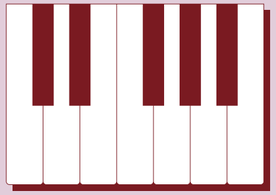
DEVELOP CLASSIFICATION MODEL

Design a machine learning model that can effectively detect and classify emotional tones from song lyrics with a high degree of accuracy



EXPAND MOOD CATEGORIES

Establish a broad range of emotional states, including happiness, sadness, anger, love, and energy levels



IMPROVE MUSIC DISCOVERY

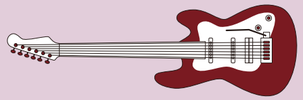
Facilitate music discovery platforms to suggest music based on emotional context, allowing users to discover music aligned with their current emotional state or desired emotional response



SCALE ACROSS LANGUAGES

Establish a system architecture that can process lyrics from different languages, taking into consideration cultural and linguistic differences in emotional expressions

EXISTING SYSTEM AND THERE LIMITATION



EXISTING SYSTEM

Spotify
• Suggests songs based on mood or playlists
-> Does NOT deeply analyze lyrics meaning

Shazam
• Identifies songs instantly
-> Does NOT have emotion or mood analysis

IBM Watson Tone Analyzer
• Detects emotions based on text
->Does NOT work well with poetic/creative lyrics

Genius
• Provides lyrics and meanings
-> Does NOT have mood classification

LIMITATIONS

Unable to understand the deeper context of lyrics

Sarcasm and hidden emotions are often missed

Only basic emotions are detected

Cannot effectively handle songs with mixed moods

Lack of advanced AI-based automation

PROPOSED SOLUTION



A multi-layered AI solution that leverages natural language processing, sentiment analysis, and deep learning to effectively classify emotional content in song lyrics:

Lyrics Preprocessing

Text data cleaning and normalization, ensuring removal of noise while retaining emotional cues and contextual meanings

Feature Extraction

Emotional keywords, metaphors, and linguistic patterns associated with various mood categories

Classification Engine

Machine learning models and algorithms applied to predict moods with confidence levels

API classification

Programmatic interface for music platforms, applications, and recommendation tools to access mood classification in real time



SYSTEM ARCHITECTURE

User input



User enters song lyrics

01

Preprocessing



→ Remove stopwords
→ Convert text to lowercase
→ Perform tokenization

02

Feature extraction



→ Convert text into numerical form (TF-IDF / Word Embedding)ode

03

Machine Processing



→ Apply Machine Learning / AI model
→ (Naive Bayes / LSTM / NLP Model)

04

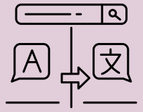
Mood classification



→ Detect emotion (Happy, Sad, Angry, etc.)

05

Output



→Display detected mood to the user

06

TECHNOLOGY USED

Frontend Development
→ HTML
→ CSS
→ JavaScript

Backend Development
→ Python

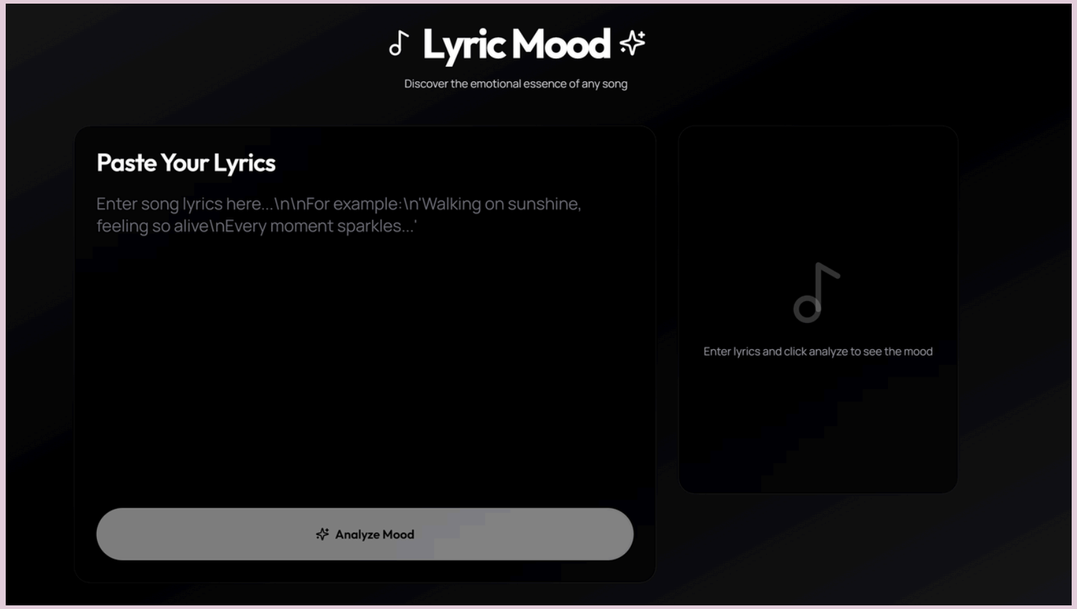
Natural Language Processing (NLP)
→ Natural Language Processing for text analysis

Machine Learning Models
→ Naive Bayes / LSTM / other NLP models

Feature Extraction Techniques
→ TF-IDF / Word Embeddings

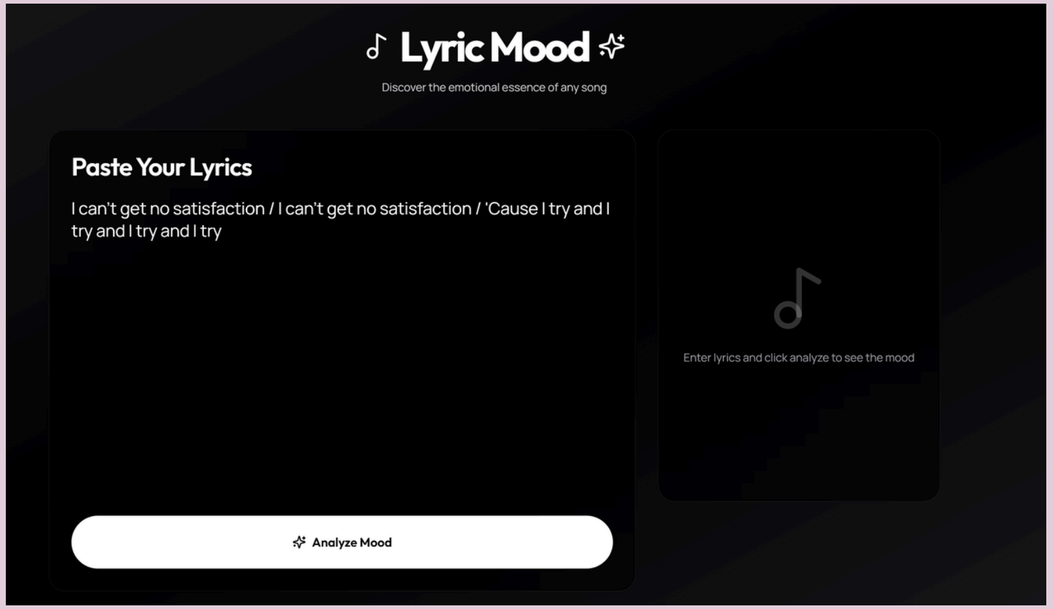


SAMPLE INPUT



OUR SITE

This is Systems UI



LYRICS INPUT

We insert lyrics

SAMPLE OUTPUT



EXPECTED OUTCOMES



Precise Mood Detection

Accurate identification of primary and secondary emotional tones within song lyrics with confidence scoring



Pattern Recognition

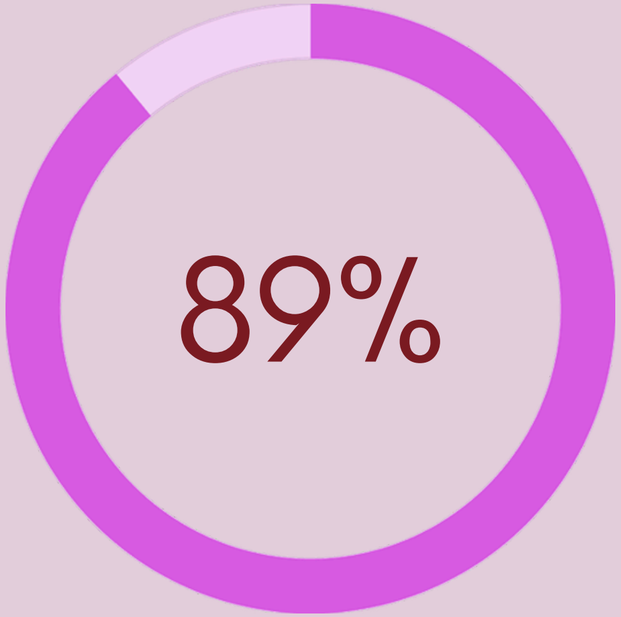
Detection of recurring emotional patterns and sentiment shifts throughout lyrical compositions



Comprehensive Analysis

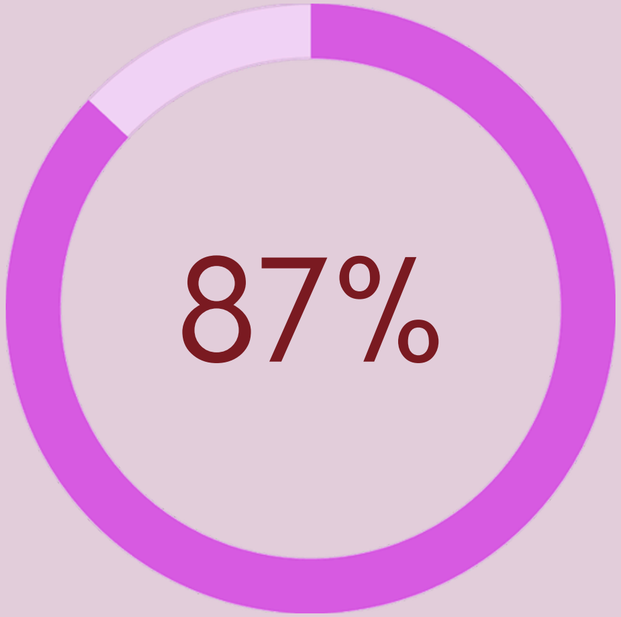
Detailed breakdown of emotional components including intensity, duration, and transitions between moods

PERFORMANCE METRICS



Genre Consistency

Accuracy maintained across rock, pop, hip-hop, and country genres



Language Support

Effective mood detection in English and Hindi lyrics

APPLICATIONS

Music Streaming Platforms

The user will be able to paste the lyrics of their chosen song and will immediately be able to know the mood of the song.



Multi-Language Support

This system can be extended to analyze the mood of the songs in Hindi and English.

Music enhances storytelling and creates memorable experiences across media

Emotion Categorization System

It will displays different emotional states and therefore, it has the potential to be used for further emotional state analysis.



Song Creation

The creators can create songs of their own choice.

FUTURE SCOPE

Mood-Based Interaction

Users can explore songs based on emotional categories and facial expressions or gesture.

Live Prediction

Instant mood detection while typing lyrics .

Language Expansion

Supports for multiple regional and global languages.

Mood-Based Meme Suggestions

It will display relevant memes based on detected mood.



FUTURE SCOPE

App Development

Convert the system into a mobile application.

Playlist Automation

Auto-generate playlists based on
detected emotions.

Voice Recognition

Analyze mood based on spoken or sung lyrics.

Industry Collaboration

Partner with music and technology companies.



REFERENCES



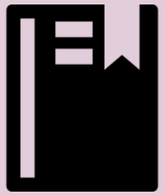
GITHUB

Several accounts and several sources.



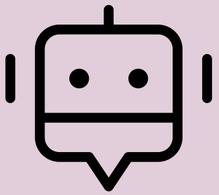
OPEN SOURCE

Source: IIIT Hyderabad
https://share.google/4KtdJSaRmEiQjCvHD



ARTICLES

Spotify Engineering Research



MODERN AI

AI chatbot

CONCLUSION

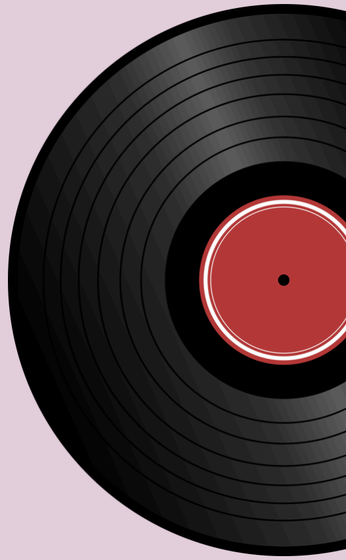
The system has successfully analyzed the song lyrics to identify the tone of the emotions in the lyrics using Natural Language Processing and Machine Learning algorithms. It processes the text data by analyzing it in terms of preprocessing, feature extraction, and prediction to identify the emotions in the text, such as happy, sad, and angry. This application has successfully provided an interface for the user in a simple manner while giving accurate output. This project has shown the potential of Al in understanding human emotions.







Larana, Inc.





THANK YOU

Thank you for your time and attention

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